

$$x(t_1) = e^{A_0(t_1-t_0)} x(t_0) + \int_{t_0}^{t_1} e^{A_0(t_1-z)} B_0 dz$$

$$= e^{A_0(t_{n_1} + \delta t_1 - t_0)} x(t_0) + \int_{t_{n_0}}^{t_{n_1} + \delta t_1} e^{A_0(t_{n_1} + \delta t_1 - \tau)} g_0 d\tau$$

$$= \frac{F_{n_0}}{e} A_0(t_{n_1} - t_{n_0}) \frac{A_0 \delta t_1}{e} x(t_0) + \frac{A_0 \delta t_1}{e} \left[\int_{t_{n_0}}^{t_{n_1}} \frac{A_0(t_{n_1} - z)}{e} g_0 dz + \right. \\ \left. + \int_{t_{n_1}}^{t_{n_1} + \delta t_1} \frac{A_0(t_{n_1} - z)}{e} g_0 dz \right]$$

$$= F_{x_0}(I + A_0 \delta t_1) x(t_0) + (I + A_0 \delta t_1)(g_{x_0} + b_0 \delta t_1)$$

$$\doteq F_{n_0} \underbrace{x(t_0)}_{x(t_0) - x_n(t_{n_0}) + x_n(t_{n_0})} + g_{n_0} + [F_{n_0} A_0 x(t_0) + A_0 g_{n_0} + b_0] \delta t_1$$

$$F_{n_0} x_n(t_{n_0}) + g_{n_0} = x_n(t_{n_1})$$

$$x(t_1) - x_n(t_{n_1}) = F_{n_0} [x(t_0) - x_n(t_{n_0})] + [F_{n_0} A_0 x(t_0) + A_0 g_{n_0} + b_0] \Delta t_1$$